

PRODUCT REQUIREMENTS DOCUMENT

RideSafe: A women-centric cab booking app

Rebuilding the ride around an AI Guardian that watches, warns, and acts - not a hidden SOS button three menus deep.

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DOCUMENT STATUS

Independent product case study, written as a working PRD - not based on any live company's internal data or roadmap.

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1. Why I'm Writing This

I started this project with a simple, uncomfortable observation: every major ride-hailing app treats safety as an afterthought. It exists, technically - there's an SOS button somewhere, usually buried two or three taps deep in a settings menu nobody opens until something has already gone wrong. For a huge number of riders, especially women traveling alone at night, that's not good enough. Safety shouldn't be a feature you have to go looking for in the worst moment of your trip.

RideSafe is my answer to that gap. It's built around one idea I kept coming back to: what if the app itself acted like a companion who was actually paying attention - noticing when a route looks wrong, checking in when it's late at night, and knowing exactly what to do the moment something feels off, without the rider having to explain it in a menu.

I didn't want to design another chatbot that answers refund questions. I wanted to design a Guardian - something that watches, warns, and can act on a rider's behalf in real time.

This document lays out what I'd build, why I believe it matters, and how I'd know if it actually worked. I'm writing it the way I'd want to read one myself - plainly, honestly, and without pretending I have more certainty than I do.

2. The Problem I See Today

I looked closely at how Uber, Ola, Lyft and Rapido currently handle safety, and the pattern was consistent across all of them: safety is present, but passive. It waits for the rider to notice something is wrong, find the right screen, and take action - all while possibly already feeling unsafe. Here's the comparison that convinced me this needed a different approach.

CAPABILITY	TODAY'S APPS	RIDESAFE
Safety surface	Hidden SOS button, several taps deep	Always-on Guardian FAB, one tap away
Night rides	Same experience as any daytime ride	Night Safety Mode with automatic live-share
Driver trust	A photo and a star rating	Live face and plate verification
Route deviation	No alert to the rider at all	AI detects it and offers real actions
Emergency	A manual call to support	One-tap SOS to trusted contacts
Gender preference	Not offered as a setting	Verified female-driver priority
Assistant	A chatbot that handles refunds	A context-aware safety companion

None of these gaps are hidden or hard to find - I found every one of them just by using the apps and reading how riders describe their own experiences online. What surprised me wasn't any single gap; it was how consistently "safety" meant "a button that exists" rather than "something that's actually looking out for you."

Female riders, especially at night, get a ride. They don't get a companion. That's the gap RideSafe is built to close.

3. Who I'm Building This For

I kept the target audience grounded in real situations rather than inventing detailed personas with made-up backstories. Three recurring situations shaped almost every decision in this PRD.

The Late-Night Commuter

Someone heading home after a late shift, a flight, or a night out - alone, tired, and more alert to their surroundings than they'd like to be. She wants the app to notice it's late and act accordingly, without her needing to remember to turn anything on herself.

The Cautious First-Timer

Someone new to a city, or simply new to trusting a stranger's car with their safety. She wants proof - a verified face, a verified plate - before she wants convenience.

The Person Waiting at Home

Not the rider herself, but a parent, partner, or friend on the other end of a shared trip link. She wants to know, at a glance and without calling, that the ride is going the way it should.

I designed for these three because they cover the whole loop - the person in the car, and the person hoping she gets home safe.

4. My Vision - Guardian AI as the Differentiator

Every feature in this document could, in theory, exist as a static toggle buried in a settings screen - exactly where today's apps already put them. What makes RideSafe different isn't any single feature; it's Guardian AI, the layer that sits on top of all of them and actually uses them on the rider's behalf.

I want to be precise about what I mean by that, because "AI safety assistant" can mean almost nothing if I don't define it clearly. Guardian AI is:

- Natural language, not menu language - a rider can say "I don't feel safe" and have SOS armed, instead of finding the right screen first.
- Context-aware - it already knows the time, the route, and which safety modes are active, so it never asks a rider to repeat information the app already has.
- Able to act, not just talk - it can actually enable Night Safety Mode, share a live location, or start an SOS workflow. It operates the app; it doesn't just describe what the app can do.
- Proactive on the rider's terms - it nudges toward safety, not toward booking another ride. If it ever starts to feel like it's selling something, I've built it wrong.

The test I kept applying while designing this: would a genuinely caring, highly competent friend do this? If the honest answer was no, the feature didn't make it in.

I also want to be clear about what Guardian AI is not. It's not a general-purpose assistant that happens to live inside a safety app - it doesn't book restaurants, answer trivia, or chat for the sake of chatting. Keeping its scope narrow is a deliberate product choice, not a technical limitation. A safety companion that tries to do everything ends up trustworthy at nothing, and trust is the entire point of this product.

5. Goals and Non-Goals

Goals

- Make safety the first thing a rider sees, not something she has to search for.
- Give Guardian AI the ability to act on real app functions, not just answer questions about them.
- Make night rides meaningfully different and safer by default, without adding extra steps for the rider.
- Replace a photo-and-rating trust model with verifiable driver identity.
- Make the people a rider trusts most - not customer support - the first line of response in an emergency.

Non-Goals

- This is not a full ride-hailing platform rebuild - booking, pricing, and driver-side tooling are out of scope for this document.
- This is not a general-purpose AI chatbot - Guardian AI's scope stays inside safety and trip context, deliberately, so it stays fast and trustworthy.
- This is not a law-enforcement integration project - RideSafe notifies trusted contacts and surfaces information quickly; it does not replace emergency services.
- This is not a redesign of every screen in the app - I focused on the ten screens and flows where safety decisions actually happen.

6. Scope - What's In, What's Out for V1

I want to be honest about sequencing. Not everything in this PRD should ship on day one, and pretending otherwise would make this a wish list instead of a plan.

INCLUDED IN V1	DELIBERATELY LATER
Home safety layer, Guardian AI FAB, Night Safety Mode	Full community safety maps (needs real ride-density data first)
Driver face and plate verification	Health Mode sub-flows beyond pregnancy and period tracking
Trusted Contacts, one-tap SOS with countdown	Code-word protection (needs voice-detection accuracy testing)
Live Map with route-deviation detection	AI notifying a specific contact by name (needs contact-intent testing)
Verified profile and ride safety timeline	Automated local-authority response integration

The V1 line isn't about which features are impressive - it's about which ones I can promise will actually work reliably before a rider's safety depends on them.

7. Core Feature Requirements

Nine capabilities make up the safety layer. I've grouped them here as an overview before going deeper on the three screen clusters that matter most.

CAPABILITY	WHAT IT DOES
Guardian AI	Natural-language safety companion that understands context and can operate app features directly.
Driver Verification	Live face and license-plate verification before and during the ride, not just a static photo.
Trusted Contacts	A ranked circle of up to five people who are notified automatically in an emergency.
Night Safety Mode	Automatically reshapes the app after sunset - darker palette, proactive nudges, safer defaults.
Route Deviation Alerts	AI monitors the live path and flags anomalies the rider might not notice herself.
Silent Audio Trigger	A discreet way to start recording or alert contacts without drawing attention.
Health Mode	Context for pregnancy, period, or medical needs that can inform how the ride is handled.
Community Safety Maps	Aggregated, anonymized signal about areas riders have flagged as lower-confidence.
Emergency SOS	One-tap, cancellable countdown that shares live location and audio with trusted contacts.

8. Feature Deep-Dive - Home, Night Safety Mode & Guardian AI

Home

The entry point has to prove the app cares about safety before the rider has even booked a ride. That means the where-to search sits alongside a primary Book Safe Ride action, Night Safety and Female Driver toggles are visible above the fold rather than in a settings menu, and a live summary of which safety layers are currently active is always on screen - no menu diving required.

Night Safety Mode

I wanted the interface itself to change when it matters most, not just offer a toggle that changes nothing visually. After sunset, the app automatically shifts to a high-contrast dark palette, Guardian AI proactively suggests enabling Trusted Watcher, and the SOS control remains the one element that never loses visual priority. The layout stays familiar - only the defaults get safer, so there's zero new cognitive load exactly when a rider has the least patience for it.

Guardian AI

The Guardian AI floating action button is reachable from anywhere in the app, always one tap away. It's the layer that turns "I don't feel safe" into an armed SOS, and it's the same layer that quietly knows the current time, route, and active safety modes so it never has to ask the rider to repeat herself.

ACCEPTANCE CRITERIA

- Night Safety and Female Driver toggles are visible on Home without scrolling, on all supported screen sizes.
- Guardian AI FAB is present and tappable from every primary screen, including mid-ride.
- Night Safety Mode activates automatically based on local sunset time, and can be manually overridden in either direction.

9. Feature Deep-Dive - Safety Center, Live Map & Route Deviation

Safety Center

Every safety feature lives on one screen, with nothing hidden behind a secondary menu. Night Safety, the Silent Audio Trigger, and Trusted Watcher sit together with Health Mode options for pregnancy, period, and medical needs. Every toggle is live - when a rider changes a setting here, Guardian AI's understanding of her state updates instantly, so it never acts on stale information.

Live Map

The Live Map is where a rider sees the ride, shares the ride, and effectively verifies the ride is going the way it should. It shows the live route with deviation alerts layered directly on top, and a one-tap share sends that same live view to trusted contacts - so the person waiting at home isn't left guessing.

Route Deviation Alerts

This is Guardian AI's quietest and, I think, most important job. It monitors the path against the expected route and warns on anomalies - not with a vague alert, but with real actions attached: contact the driver, notify a trusted contact, or arm SOS. A deviation isn't automatically an emergency, so the response has to offer options, not force one.

ACCEPTANCE CRITERIA

- A route deviation beyond a defined threshold surfaces an in-app alert within seconds of detection.
- Every Safety Center toggle change is reflected in Guardian AI's context within the same session, with no refresh required.
- Live Map sharing works with a single tap and requires no additional confirmation once a trusted contact is selected.

10. Feature Deep-Dive - Trusted Contacts, SOS & Verified Profile

Trusted Contacts

These are the people who get called before the police do - by design. A ranked circle of up to five watchers means SOS doesn't have to guess who to notify first. When SOS triggers, it auto-dials the top contact and shares live location with the full circle, and Guardian AI can notify a specific person by name if a rider asks for one directly.

SOS

SOS needs to work under pressure, which means it has to be fast without being reckless. A five-second, cancellable countdown gives a rider a moment to stop an accidental trigger, after which it auto-shares live location and audio, notifies every trusted contact, and surfaces local emergency information. It can be triggered by the button, by a code-word, or directly through Guardian AI recognizing intent in what a rider says.

Verified Profile

Trust has to run in both directions. A rider's profile shows her own ID and face-verification status, a ride history with a visible safety timeline, and her stated preference for female drivers - all in one place, so trust isn't something she has to re-establish every time she opens the app.

ACCEPTANCE CRITERIA

- SOS countdown is cancellable for the full five seconds and clearly communicates time remaining.
- All active trusted contacts receive a live-location share within moments of SOS triggering, with no manual step required.
- Guardian AI can correctly identify a named trusted contact from natural spoken or typed input at least as reliably as a manual contact search.

11. User Stories, Edge Cases, Dependencies and Risks

Key user stories

- As a late-night commuter, I want the app to notice it's dark out and offer to turn on extra protection, so I don't have to remember to do it myself.
- As a cautious first-time rider, I want to see a live face match against my driver before I get in, so I'm not trusting a photo that could be years old.
- As someone waiting at home, I want a link that shows me the live route without needing the rider to call me, so I know she's safe without interrupting her.
- As a rider who suddenly feels unsafe, I want to say so out loud or in text and have the app act immediately, without hunting for a button.

Important edge cases

- Rider has no trusted contacts added yet - SOS should still notify local emergency information and clearly prompt her to add contacts afterward, not fail silently.
- Route deviation happens for a legitimate reason, like a road closure - the alert should offer a quick "this is expected" dismissal so real anomalies don't get lost in false positives.
- Rider triggers SOS accidentally - the cancellable countdown exists exactly for this, and canceling should require no more than one tap.
- Driver's live face verification fails to match at pickup - the rider needs an immediate, clear option to cancel the ride safely before getting in.
- Guardian AI misunderstands a natural-language request - it should default to showing the closest matching safety action for confirmation, never silently doing nothing.

Dependencies

Real-time driver-side location and identity data, a natural-language understanding model scoped tightly to safety intents, telephony and SMS infrastructure for trusted-contact notification, and map and routing data reliable enough to distinguish a genuine deviation from ordinary traffic rerouting.

Risks

- False-positive route deviations could train riders to ignore real alerts - the detection threshold needs careful tuning before wide rollout, not just a reasonable-sounding default.
- Face verification raises real privacy and data-handling questions that deserve their own dedicated review, not a footnote in this document.
- Guardian AI acting on misheard or misread intent could trigger an unwanted SOS - every AI-triggered action above a certain severity should show a brief, cancellable confirmation, never fire silently.
- Verified female-driver priority depends on driver-side supply in a given city, and I don't have that data - this could promise something the marketplace can't yet deliver everywhere.

- Battery and connectivity loss mid-ride would silence route-deviation monitoring and live sharing at exactly the wrong moment - the app needs a clear, rider-visible signal when this protection has gone dark, not a silent failure.

None of these risks are reasons to hold the product back - they're reasons to sequence it carefully, which is exactly what the rollout plan on the next page is built around.

12. How I'd Measure Success, Rollout & What I'd Do Next

How I'd measure this

I don't have production data to build real baselines from, so I'm not going to invent numbers here. What I can do is name the signals I'd actually watch:

- Percentage of night rides with Night Safety Mode active, and how many of those were AI-suggested versus manually turned on.
- Time from route-deviation detection to rider action, and the false-positive rate on those alerts.
- SOS trigger-to-first-contact-notified time, end to end.
- Guardian AI intent-recognition accuracy on safety-specific phrasing, tracked separately from general chatbot metrics.
- Growth in trusted-contact circles per rider over the first month, as a proxy for whether the feature feels worth setting up.

Rollout plan

I'd start in a small number of cities with strong night-ride volume and enough verified female drivers to make the gender-preference promise real, not aspirational. Guardian AI would launch with a narrow, well-tested set of safety intents rather than broad natural-language coverage, and I'd expand its scope only as accuracy data justified it.

What I'd do next

- Talk to riders directly, especially the late-night commuter segment, before finalizing the route-deviation threshold.
- Pressure-test Guardian AI's intent recognition against real, messy natural language, not clean scripted phrases.
- Bring in a privacy and safety specialist to review face verification and SOS data handling before wide rollout.
- Prototype the SOS and route-deviation flows first, since they carry the most real-world consequence if they're wrong.

What I learned writing this

The feature that felt most technically interesting to design - Guardian AI - only works because of everything underneath it: verified drivers, a trusted circle, a route the app actually understands. Safety isn't one clever

feature bolted onto a ride-hailing app. It's the app quietly reorganizing itself around the person in the back seat, at exactly the moments she needs it most.